

[Domain:] All people $S(x)$: x is a student of your class.

$$\exists x(S(x) \wedge \neg C(x))$$

d) A student in your class has been in a movie.

[Domain:] Students in your class $M(x)$: x has been in a movie

$$\exists x M(x)$$

[Domain:] All people $S(x)$: x is a student of your class.

$$\exists x(S(x) \wedge M(x))$$

e) No student in your class has taken a course in logic programming.

[Domain:] Students in your class $LP(x)$: x has taken a course in logic programming

$$\forall x \neg LP(x)$$

[Domain:] All people $S(x)$: x is a student of your class.

$$\forall x(S(x) \rightarrow \neg LP(x))$$

25) Translate each of these statements into logical expressions using predicates, quantifiers and logical connectives.

~~Everyone is perfect~~ Domain: All people. $F(x)$: x is my friend

$P(x)$: x is perfect

a) No one is perfect: $\forall x \neg P(x)$ ✓

b) Not everyone is perfect: $\exists x \neg P(x)$ ✓

c) All your friends are perfect: $\forall x(F(x) \rightarrow P(x))$ ✓

d) At least one of your friends is perfect: $\exists x(F(x) \wedge P(x))$ ✓

e) Everyone is your friend and is perfect: $\forall x(F(x) \wedge P(x))$ ✓

f) Not everybody is your friend or someone is not perfect:

$$\exists x(\neg F(x)) \vee \exists x(\neg P(x))$$

27) Translate each of these statements into logical expressions in three different ways by varying the domain and by using predicates with one and with two variables.

$C(x)$: x is a student of your class $S(x)$: x is a student of your school.